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Case 1-2022: A 67-Year-Old Man with Motor Neuron Disease and Odd Behaviors during Sleep

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PRESENTATION OF CASE

Dr. Suma Babu: A 67-year-old man was evaluated in the sleep clinic at this hospital because of odd behaviors during sleep, daytime sleepiness, and apnea detected on a home sleep study.

Three years before the current evaluation, the patient's wife noticed that the patient had vocalizations, including talking and yelling, while sleeping. He also had thrashing movements while sleeping; he fell out of bed more than once and inadvertently hit his wife during these movements.

Two years before the current evaluation, the patient's wife noticed that the patient had prolonged episodes of daytime sleepiness. During the episodes, he appeared to fall asleep while sitting up and was unresponsive to shaking and loud voices; several hours later, he returned to normal interaction without any intervention. The daytime sleepiness was intermittent and fluctuated in severity.

Four months before the current evaluation, the patient fell asleep while washing dishes. This resulted in a fall and unstable C1 spinal fracture, which required urgent surgical repair. After the surgical repair, melatonin and trazodone were prescribed for insomnia, and abnormal movements during sleep abated. However, vocalizations during sleep — including singing, talking, and expression of fear and anger in the context of upsetting dreams — continued and were most prominent in the early morning hours.

The patient also had a motor neurologic syndrome that had started before the onset of sleep symptoms, approximately 3.5 years before the current evaluation. Symptoms included progressive gait imbalance with falls, hand clumsiness, hoarse voice, and dysphagia. Serial neurologic examinations, performed over a 3-year period by multiple neuromuscular specialists, revealed slow progression of generalized fasciculations and the development of bulbar hyperreflexia, postural instability, and hypophonia, although strength in the arms and legs remained intact. Serial electromyography revealed progressive and widespread involvement of muscles showing chronic reinnervation changes, including eventual involvement of craniobulbar muscles (tongue and masseter muscles).

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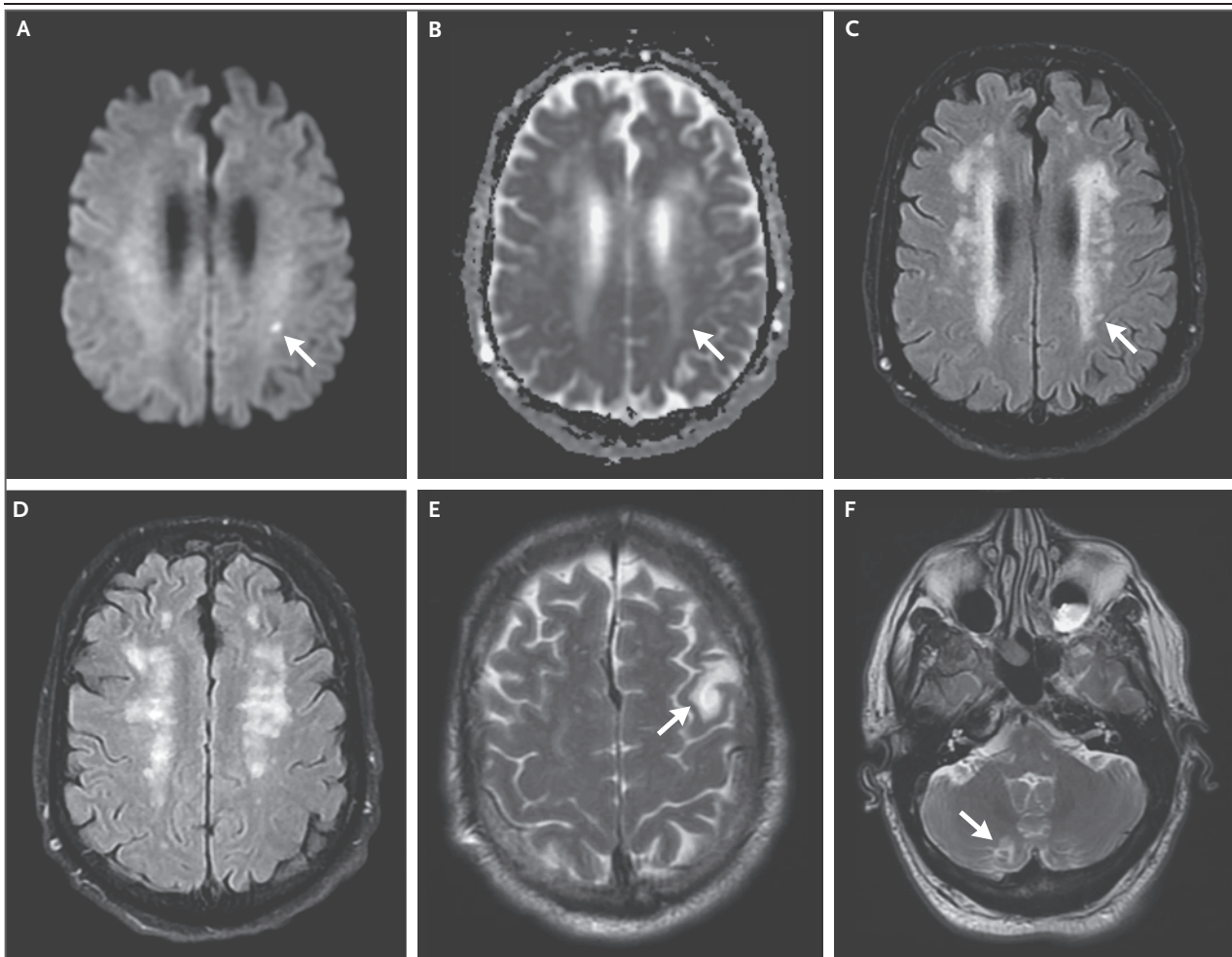


Figure 1. MRI of the Head.

MRI of the head was performed 4 months before the current evaluation. Diffusion-weighted images (Panels A and B) and a fluid-attenuated inversion recovery (FLAIR) image (Panel C) show a punctate subacute infarct in the left posterior centrum semiovale (arrows). FLAIR images (Panels C and D) show nonspecific moderate-to-severe background white-matter changes. T2-weighted images (Panels E and F) show chronic infarcts in the left middle frontal gyrus and right cerebellar hemisphere (arrows).

Dr. Brian Zhao: One year before the current evaluation, magnetic resonance imaging (MRI) of the cervical spine revealed moderate neural foraminal narrowing at the level of C3–C4 bilaterally and no central canal stenosis. Four months before the current evaluation, MRI of the head revealed a punctate subacute infarct in the left posterior centrum semiovale, chronic infarcts in the left middle frontal gyrus and right cerebellum, and nonspecific moderate-to-severe white-matter changes (Fig. 1).

Dr. Babu: Blood tests for paraneoplastic, nutritional, and paraproteinemic causes of the patient's neurologic syndrome were unrevealing. A

diagnosis of slowly progressive motor neuron disease was made, and therapy with riluzole was begun. A videofluoroscopic swallowing study revealed severe oropharyngeal dysphagia with laryngeal penetration and silent aspiration; a gastrostomy tube was placed, and tube feeding was initiated.

The patient underwent a home sleep-apnea test, which showed sleep-disordered breathing with an apnea-hypopnea index of 20 events per hour; hypoxemia was present at rest while he was breathing ambient air. Severe oxygen desaturation occurred with frequent respiratory events while he was asleep in the supine position.

The patient was then referred to the sleep clinic for an initial evaluation.

On evaluation in the sleep clinic, the patient reported that obstructive sleep apnea had been diagnosed on the basis of a home sleep study performed 1 year earlier. Multiple attempts at treatment with continuous positive airway pressure (CPAP) had been unsuccessful; there had been difficulty in finding a mask that fit comfortably.

In addition to the sleep and motor neurologic syndrome described previously, the patient's symptoms included snoring, difficulty with sleep initiation, forgetfulness, poor judgment regarding his ability to safely complete physical tasks, urinary urgency and frequency, and night sweats. He had a history of type 2 diabetes mellitus, which was well controlled with insulin and had been complicated by mild, stable, length-dependent sensory polyneuropathy; other history included hyperlipidemia, a heterozygous factor V Leiden mutation, depression, orthostatic hypotension, and gastroesophageal reflux disease. Medications included alprazolam, bupropion, fenofibrate, insulin, melatonin, metformin, riluzole, sertraline, and trazodone; there were no known drug allergies. The patient lived in New England with his wife. He was a retired engineer. He drank two alcoholic beverages per day. He did not smoke tobacco or use illicit drugs. His mother had died from an unknown cause at an elderly age; his father had died from trauma. One of his brothers had cerebrovascular disease, and another had schizophrenia.

On examination, the temperature was 36.6°C, the blood pressure 112/60 mm Hg, the pulse 83 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 24.2. The patient was awake, alert, and oriented, although his attention was somewhat diminished. He could recall two words out of three. Naming and repetition were intact. The pupils were equally reactive to light and accommodation. Extraocular movements were intact, and visual fields were full. Facial sensation and strength were normal. Speech was hypophonic and mildly slurred. Elevation of the palate was symmetric, and movement of the tongue was normal. Shoulder shrug was symmetric. Strength was 5/5 in all muscle groups.

Deep-tendon reflexes were brisk in the arms, with mild spasticity in the legs. Plantar responses were normal. Vibratory sensation in the legs was mildly diminished. Results of finger-to-nose and heel-to-shin testing were normal. There were mild myoclonic movements in both hands. Alternating movements were mildly clumsy. Tone was slightly increased in the legs. The gait was on a narrow base and was unsteady; the patient was unable to walk with a tandem gait. The Romberg test showed truncal titubation.

A sleep study to determine the appropriate positive airway pressure (PAP) therapy was performed. In CPAP mode, there was good control of respiratory events. Electromyographic signals revealed elevated muscle tone during rapid-eye-movement (REM) sleep, but there were no abnormal behaviors.

Two weeks later, during a follow-up visit in the neuromuscular clinic, the patient became unresponsive to sternal rub; the vital signs and blood glucose level were normal. He was transported to the emergency department, where he returned to a normal level of consciousness without intervention. Electroencephalography (EEG) revealed frequent intermittent diffuse polymorphic delta slowing of the background, often with bifrontal predominance, but showed no epileptiform abnormalities. The patient was fitted with a mask for CPAP, and treatment was initiated. Several weeks after the initiation of treatment, download of CPAP data revealed good effectiveness and adherence.

However, episodes of unresponsiveness increased in frequency and lasted up to 3 hours at a time. During the episodes, the patient's wife was unable to arouse him despite performing a sternal rub. Modafinil was prescribed, but the episodes did not abate. The patient was admitted to the epilepsy monitoring unit for long-term monitoring and EEG. During the admission, two episodes of unresponsiveness occurred, during which the patient had a normal blood pressure, heart rate, and glucose level and EEG revealed findings consistent with sleep.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Aleksandar Videnovic: I evaluated this patient when he was referred to the sleep clinic, and I am aware of the final diagnosis in this case.

After I was asked to see this 67-year-old man with progressive motor neuron disease, my initial thoughts were centered around the detection of sleep-disordered breathing on a home sleep-apnea test, along with his previous diagnosis of sleep apnea in the context of motor neuron disease and his history of unsuccessful treatment with CPAP.

SLEEP-DISORDERED BREATHING

This patient's symptoms of snoring, breathing pauses during sleep, and daytime sleepiness are supportive of the diagnosis of sleep-disordered breathing. The findings on an initial home sleep-apnea test had led to the diagnosis of obstructive sleep apnea. The timely diagnosis and adequate treatment of sleep-disordered breathing are very important for overall health and quality of life. The standard treatment is PAP, with CPAP being used most frequently. However, patients with neuromuscular disorders may have clinically significant hypoventilation at night, in addition to apnea or hypopnea, and may receive bilevel PAP treatment.¹

Therefore, during the initial evaluation, my first question was whether the unsuccessful treatment with CPAP was related to coexisting central sleep apnea that had not been detected on the initial home sleep-apnea test or to a need for bilevel PAP, which would enable oxygenation overnight. I recommended that the patient undergo a study in our sleep laboratory to determine the appropriate treatment for his sleep-disordered breathing. The complete sleep study also allowed us to gather information about EEG activity during sleep, sleep stages, arousals from sleep, muscle activity, and unusual behaviors during sleep; a home sleep-apnea test cannot provide such a detailed characterization of sleep.

ABNORMAL BEHAVIORS DURING SLEEP

The patient had additional sleep-related symptoms, specifically difficulty with sleep initiation and abnormal behaviors while asleep. Difficulty with sleep initiation may be due to sleep-onset insomnia, which has many causes. The patient's abnormal behaviors during sleep included vocalizations and movements, along with episodes of falling out of bed during sleep. These behaviors are suggestive of parasomnias, a group of sleep disorders characterized by abnormal behaviors during sleep.

Parasomnias

Parasomnias are divided into non-REM (NREM) and REM forms, depending on whether the behaviors emanate from NREM or REM sleep.^{2,3} Certain clinical characteristics can help to differentiate between these two forms. For example, NREM parasomnias — such as sleepwalking, sleep terrors, and confusional arousals — often occur in younger patients, mainly happen during the first half of the night, and are associated with a lack of clear dream recall. During NREM parasomnias, the patient is often difficult to arouse, and the associated movements have a rather normal appearance. In this patient, the abnormal behaviors mainly happened during the second half of the night and were characterized by vocalizations and movements that indicated emotions of fear and anger in the context of upsetting dreams. These episodes suggested that the patient was acting out his dreams, which is a hallmark feature of a REM parasomnia known as REM sleep behavior disorder.

REM Sleep Behavior Disorder

The loss of muscle atonia that accompanies normal REM sleep is the neurophysiologic signature of REM sleep behavior disorder. Therefore, a sleep study is needed to establish the diagnosis; this was another reason to perform the sleep study in this patient.

It is important to emphasize that not everything that looks and sounds like REM sleep behavior disorder is indeed that condition.⁴ The differential diagnosis includes nightmares, seizures, psychogenic disorders, malingering, and sleep-disordered breathing. Nightmares are characterized by unpleasant dream imagery during REM sleep. Dream recall is usually very detailed, sleep may not be interrupted, and vocal and motor behaviors are absent. Seizures, specifically those originating from the frontal or temporal lobes, can be differentiated from REM sleep behavior disorder on the basis of the presence of stereotypic tonic and dystonic posturing. Video polysomnography with an extended EEG montage should be performed when epileptic seizures are suspected.

Psychogenic disorders occur on a background of overnight wakefulness. Psychological testing may be helpful in distinguishing these disorders from REM sleep behavior disorder. Malingering can manifest with a variety of unusual behaviors at

night. Video polysomnography is critical in documenting whether the events are occurring during wake time and in confirming the absence of characteristic findings of REM sleep behavior disorder. Complex behaviors that resemble events of REM sleep behavior disorder can occur during arousal caused by respiratory events in the context of sleep apnea. These pseudo-events of REM sleep behavior disorder resolve with the use of CPAP.

The patient underwent a sleep study in CPAP mode, and the sleep-disordered breathing abated. He was subsequently fitted with a mask for CPAP, and treatment was initiated. The sleep study also revealed increased muscle activity during REM sleep, a finding that confirms the diagnosis of REM sleep behavior disorder. He was treated with melatonin, one of a few medications used to treat this condition, which resulted in decreased dream enactment.

EXCESSIVE SLEEPINESS

The patient had fallen asleep while washing dishes. He had several episodes of unresponsiveness, which were preceded by sleepiness. During these episodes, his wife was unable to arouse him. He would eventually, sometimes after many hours, wake up and be back to normal. There was no mention of involuntary movements, incontinence, or signs of cardiorespiratory distress during the episodes.

Despite treatment with CPAP, the patient continued to have excessive sleepiness. Because this residual sleepiness could have been due to poor adherence to CPAP or poor effectiveness of CPAP, we checked his CPAP data, which were satisfactory. For the residual sleepiness, he was treated with modafinil, but he did not have a clinically significant increase in alertness.

The differential diagnosis of these episodes of unresponsiveness includes seizure disorder, syncope, arrhythmia, a cerebrovascular event, and substance use. Seizures were unlikely to explain these episodes because there was no motor activity and no confusion on arousal. Syncope would not be associated with prolonged unresponsiveness. There was no history of findings suggestive of cardiac or cerebrovascular abnormalities. Although sleepiness is the defining phenotype of narcolepsy, there were no other clinical features suggestive of narcolepsy, and such prolonged episodes of unresponsiveness would be atypical.

Because the episodes of unresponsiveness were increasing in frequency, the patient was admitted to the neurology unit of this hospital for long-term monitoring. During the admission, two episodes occurred. Corresponding EEG monitoring revealed findings consistent with sleep, mostly stage 2 (N2) and stage 3 (N3) NREM sleep, without evidence of epileptiform activity.

In summary, this patient's sleep-wake phenotype consisted of sleep-disordered breathing that was responsive to CPAP treatment, behaviors of REM sleep behavior disorder that were responsive to melatonin treatment, and excessive sleepiness with prolonged episodes of unresponsiveness (sleep) that did not abate after modafinil treatment.

OTHER SYMPTOMS

As I learned more about the patient's sleep-wake phenotype, I tried to integrate his symptoms of poor sleep and poor alertness with his other symptoms, specifically the motor neuron phenotype (predominantly bulbar features with dysarthria and dysphagia), the autonomic dysfunction, and the decline in balance. Sleep-disordered breathing is commonly associated with motor neuron disease, but REM sleep behavior disorder and autonomic symptoms are not. In addition, his imbalance was unlikely to be related to his motor neuron disease, given that he had normal strength in his arms and legs.

Neurodegenerative Disorders

REM sleep behavior disorder has been linked to neurodegenerative disorders and is now recognized as a prodromal stage of three evolving synucleinopathies: Parkinson's disease, dementia with Lewy bodies, and multiple-system atrophy.⁵ REM sleep behavior disorder is exceedingly rare in patients with other neurodegenerative disorders. The combination of this patient's signs and symptoms did not fit well with any of the three neurodegenerative disorders associated with REM sleep behavior disorder. Parkinsonism is a hallmark of these disorders and was not a feature of this patient's presentation. Although balance can be affected in any of these disorders, a balance abnormality is unlikely to be present in the early stages and to be rapidly progressive, except in patients with multiple-system atrophy of the cerebellar type. Patients with this condition have an ataxic gait, which was not present

in this patient. Autonomic dysfunction is common in patients with multiple-system atrophy, but bulbar symptoms are not. Profound lapses in consciousness from falling asleep are not characteristic of any of the three disorders. The question remained of whether we could fit the patient's sleep and alertness problems and other neurologic manifestations into a unifying diagnosis.

Anti-IgLON5 Disease

As I considered this question, a disorder that had been described relatively recently came to mind: anti-IgLON5 disease. This disorder is associated with autoantibodies to IgLON5, a neural cell adhesion molecule of unknown function.⁶ The disorder affects men and women equally and usually develops at 50 to 60 years of age.

Anti-IgLON5 disease is characterized by a distinctive sleep phenotype and a wide range of associated neurologic symptoms. Sleep phenotypes of the disorder encompass poorly consolidated sleep, NREM parasomnia with undifferentiated or poorly differentiated NREM sleep early in the sleep episode with vocalizations and movements, sleep-disordered breathing, REM sleep behavior disorder, and stridor.⁷ Daytime sleepiness may be present. Other symptoms include dysarthria, dysphagia, gait instability, chorea, eye movement abnormalities, cognitive changes, fasciculations, and myoclonus. Several clinical phenotypes of anti-IgLON5 disease have been recognized: a predominantly sleep-related disorder, a bulbar syndrome with or without chronic reinnervation changes in arm and leg muscles on electromyography, a progressive supranuclear palsy–like syndrome, a cognitive syndrome, and hyperexcitability with cramps, myoclonus, and fasciculations.⁷⁻⁹

The clinical heterogeneity of anti-IgLON5 disease, the rarity of the disease, and the slow progression over a period of months or years are factors that frequently contribute to a delay in diagnosis. The median time from symptom onset to diagnosis is 2.5 years. The clinical heterogeneity can also lead to misdiagnosis with one of several conditions, such as isolated sleep-disordered breathing, REM sleep behavior disorder, motor neuron disease, myasthenia gravis, progressive supranuclear palsy, multiple-system atrophy, Huntington's disease, or stiff-person

syndrome. Because the diagnosis of anti-IgLON5 disease is usually delayed, the symptoms are often quite prominent at the time of diagnosis. Death occurs within 3 years after diagnosis in more than 50% of patients.

Current knowledge about treatment effectiveness is limited. Most patients are treated with immunotherapy including glucocorticoids, immunoglobulins, plasma exchange, rituximab, cyclophosphamide, azathioprine, mycophenolate mofetil, or a combination of these therapies.¹⁰ The reported response rate to immunotherapy ranges from 10 to 70%. Combination therapy seems to be more effective than monotherapy.

This patient had many of the typical sleep symptoms reported in patients with anti-IgLON5 disease, as well as a combination of neurologic symptoms reported in patients with clinical subtypes of the disease. To establish this diagnosis, testing for IgLON5 autoantibodies was performed.

DR. ALEKSANDAR VIDENOVIC'S DIAGNOSIS

Anti-IgLON5 disease.

DIAGNOSTIC TESTING

Dr. Jenny J. Linnoila: Initially, serum paraneoplastic antibody panel testing was performed in a neuroimmunology reference laboratory. The testing was negative; at that time, tests for IgLON5 autoantibodies were not commercially available.

Since then, tests for IgLON5 autoantibodies have become commercially available. These tests have equal sensitivity when performed on serum and on cerebrospinal fluid (CSF).⁸ This is not the case with tests for all neural autoantibodies. For example, testing for anti-N-methyl-D-aspartate receptor antibodies, which are associated with the most common form of autoimmune encephalitis, is more sensitive with CSF than with serum,¹¹ whereas testing for anti-leucine-rich, glioma-inactivated 1 antibodies, which are associated with the second most common form of autoimmune encephalitis, is more sensitive with serum than with CSF.¹²

In general, when an autoimmune neurologic condition is suspected, good practice involves testing of both serum and CSF with autoanti-

body panels. Conditions associated with different neural autoantibodies can have overlapping symptoms, especially early on; with the continually broadening array of neural autoantibodies included on panels, it is increasingly difficult to predict ahead of time which antibodies could be detected. Also, it is important to note that many newly identified neural autoantibodies, such as IgLON5 autoantibodies, are not included on paraneoplastic panels (which should therefore be avoided), but they appear on newer autoimmune panels, including panels of autoimmune encephalitis–associated antibodies.

In this patient, because there was a high clinical suspicion for anti-IgLON5 disease, testing for IgLON5 autoantibodies was performed on a research basis (given that commercial tests were not available at the time), with the serum specimen from the initial paraneoplastic antibody panel testing. IgLON5 autoantibodies were first identified on a tissue-based indirect immunofluorescence assay. In this assay, the patient's serum sample was applied to a composite of mouse brain, gut, and kidney tissue. Tissue-binding anti-IgLON5 IgG in the sample was visualized with the use of a fluorescently labeled secondary antibody to human IgG. In the brain, anti-IgLON5 IgG is known to bind to the cerebellum (to the granular layer more than the molecular layer), midbrain, and thalamus and binds less strongly to the hippocampus and cortex.⁸ It is important to assess the binding pattern on gut and kidney tissue, as well, because anti-IgLON5 IgG also binds to intestinal smooth muscle and renal glomeruli.

Although this tissue-binding pattern is highly suggestive of the presence of IgLON5 autoantibodies, the diagnosis of anti-IgLON5 disease must be confirmed by testing the sample with a cell-based assay specific for IgLON5. In this assay, the patient's serum sample was applied to human embryonic kidney 293 (HEK293) cells expressing the IgLON5 protein. Because IgLON5 is a cell adhesion molecule, when it is transfected into HEK293 cells, it is expressed on the cell surface. Bound anti-IgLON5 IgG was visualized with the use of a fluorescently labeled secondary antibody to human IgG. Both the tissue-based indirect immunofluorescence assay and the cell-based assay of the patient's serum sample were positive for anti-IgLON5 IgG, and the

patient received a diagnosis of anti-IgLON5 IgG–associated neurologic disorder, or anti-IgLON5 disease.

LABORATORY DIAGNOSIS

Anti-IgLON5 IgG–associated neurologic disorder.

FOLLOW-UP

Dr. Haatem M. Reda: After the diagnosis of anti-IgLON5 disease was established, the patient was treated with intravenous methylprednisolone (1000 mg daily) for 5 days. The patient's illness coincided with the beginning of the coronavirus disease 2019 pandemic, which delayed further treatment. He received intravenous immune globulin (2 g per kilogram of body weight monthly) for 2 months, until two induction infusions of intravenous rituximab (1000 mg 2 weeks apart) could be arranged.¹³

Symptoms of REM sleep behavior disorder and sleep-disordered breathing abated during the first 4 months after treatment was initiated. Approximately 1 month before the first maintenance dose of rituximab, which had been planned for 6 months after the second induction dose, was to be administered, the patient's wife reported that the patient had worsening symptoms of REM sleep behavior disorder. The interval between rituximab doses was shortened to 4 months, and the symptoms abated. Episodes of excessive daytime sleepiness resolved after the initiation of rituximab therapy.

For persistent gait unsteadiness, the patient has undergone physical therapy and still requires either a walker or the assistance of at least one person for stability. Despite precautions, however, he continues to fall approximately once per week at home. He continues to have cognitive impairment, with abnormalities on the recall and orientation domains of the Montreal Cognitive Assessment screening tool (score, 22; range, 0 to 30, with lower scores indicating greater cognitive impairment). Oropharyngeal dysphagia remains a substantial problem, for which the patient requires ongoing use of a gastrostomy tube; he has had multiple hospitalizations for recurrent complications of aspiration. He continues to have speech impairment and to undergo speech therapy. Limited knowledge about the

effects of immunotherapy in patients with anti-IgLON5 disease prevents us from making conclusions regarding the selective effectiveness of the treatment that this patient has received. Irreversible neurodegeneration embedded within anti-IgLON5 disease may underlie this limited response to immunotherapy.

FINAL DIAGNOSIS

Anti-IgLON5 IgG–associated neurologic disorder.

This case was presented at Neurology Grand Rounds.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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